

Fleet-wide stability masks change in the Maine lobster fishery (2008–2022)

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Abstract

For most of the past few decades, landings in the American lobster (*Homarus americanus*) fishery in Maine have been increasing, but a recent downturn in catch suggests the fishery may be at an inflection point. Drawing on multiple datasets associated with the fishery, we use this period of transition to review fleet dynamics in the fishery by analyzing how fishing effort has changed through time (2008–2022). When possible, age, gender, geography, and scale of fishing operation are considered to delimit intra-fleet differences. The results of this review reveal large-scale changes in intra-fleet dynamics that help to explain how there has been the appearance of fleet-wide stability for most of the 15-year study period despite mounting socioeconomic and environmental stressors. Changes in intra-fleet dynamics are most evident across geography and scale of fishing operation. In addition, this study finds that prior research has overestimated a key metric of fishing effort in the Maine lobster fishery by an order of magnitude. This latter insight bears significance because the lobster industry is under mounting pressure to reduce risk of gear interactions with large marine mammals, and future management decisions will likely hinge on estimates about fishing effort and the probability of marine mammal interactions. Continued efforts to understand fishing fleet dynamics and how they differ among segments of the fishery are vital to making well-informed policy decisions in the face of change, including the iconic Maine lobster fishery.

Keywords: adaptation, catch per unit effort; fisheries management; lobster; Maine

Introduction

Integrating socioeconomic information into fisheries management

Efforts to manage commercial fisheries have largely focused on understanding the biophysical and environmental dynamics of fish stocks and the natural environments within which they exist. The information to inform this work has traditionally been gained by focusing on biological indicators of stock abundance derived from fishery dependent and independent data such as landings, catch rates, and sampling surveys (e.g. Maunder et al. 2006, Cook et al. 2020). Socioeconomic information related to fleet behavior, including fishery participation, seasonality (i.e. timing and duration of fishing), and catch per unit effort complement biological data (e.g. Felthoven and Kasperski 2013) by providing critically important insights into how contextual factors (e.g. vessel operating costs, regulatory change, and financial health) are shaping fisheries and the wellbeing and resilience of harvesters. Despite the importance of this information, socioeconomic dimensions of fisheries remain chronically understudied (Hicks et al. 2016), including in data-rich fisheries such as the American lobster (*Homarus americanus*) fishery in the northeast USA.

In recent years, this socioeconomic data gap has been a point of emphasis for the American Lobster Stock Assessment Subcommittee (SAS) of the Atlantic States Marine

Fisheries Commission (ASMFC 2020), which is responsible for developing stock assessments for the lobster fishery in the USA. Recognizing this deficiency, the SAS has recommended the creation of an indicator (“Fishery Target”) that considers the socioeconomic resilience of the fishery and has also called for additional research to refine managers’ understanding of the socioeconomic dynamics of the fleet.

Interest in the social and economic dimensions of the lobster fishery and the wellbeing, adaptive capacity, and resilience of the people who depend on it is being driven by recognition that the fishery, such as many others around the world, is experiencing “drastic changes” and these changes create risk and vulnerability for harvesters (ASMFC 2020, pg. 3). Not only have landings decreased significantly from their peak in 2016 after nearly two decades of near-continuous growth (ACCSP 2023), but the lobster fishery has also been confronted by a number of other socioeconomic and environmental stresses associated with shifting-settlement patterns and decreasing productivity of lobster (Casey et al. 2023, McManus et al. 2023), declining bait availability (Stoll et al. 2022), and global trade disruptions and market turbulence (Stoll et al. 2018). As we discuss in the subsequent section, these issues have been further compounded by questions surrounding the potential risk that lobster gear poses to North Atlantic right whales (NARW) (*Eubalaena glacialis*).

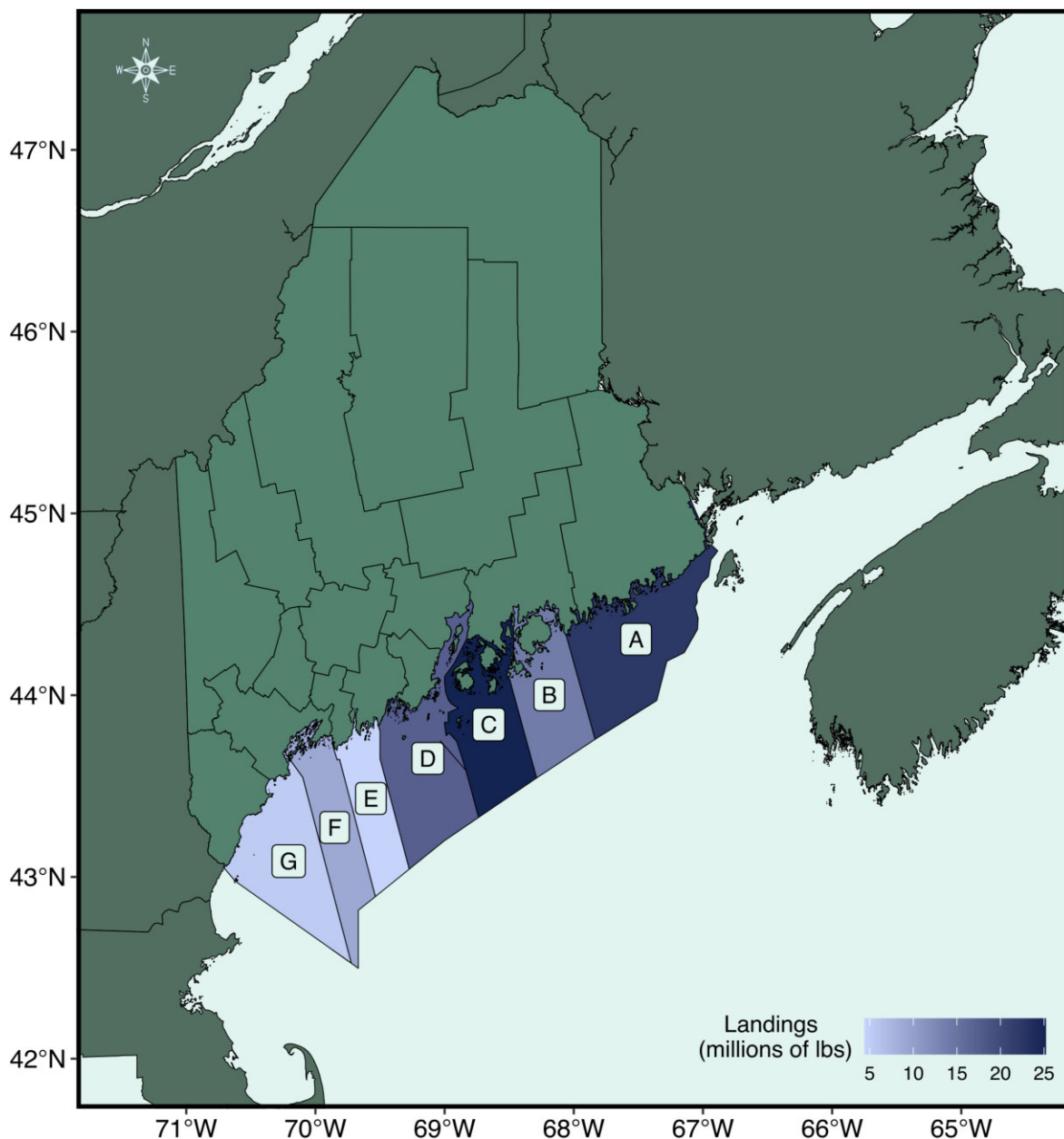


Figure 1. Map of the Gulf of Maine showing the seven Maine Lobster Management Zones (G–A) color coded by 2022 lobster landings. Federal Lobster Conservation Management Areas are denoted in blue. Note: Area 1 includes all of the Lobster Management Zones.

Acknowledging the diverse stressors facing the lobster fishery and aiming to support the integration of socioeconomic information into future decision-making, this article provides a review of fishing effort in the Maine lobster fleet using licensing, harvester, and dealer data spanning a 15-year period from 2008 to 2022. In conducting our review, we ask: How has fishing effort in the lobster fishery changed during a time period characterized by increasingly diverse stressors? We begin by providing a brief description of the lobster fishery and explain why understanding fishing effort is salient to ongoing management efforts. Next, we provide a summary of the data used in our study and describe the metrics we use to analyze fishing effort. We then describe fishing effort patterns across and within the fleet over the last 15 years. As fisheries face mounting environmental and socioeconomic stressors, understanding the ways in which fishing fleets are adapting is vital to

making effective management decisions that support healthy fisheries and resilient fishing fleets.

Methods

Study system

The American lobster fishery, spanning the northeast United States (US) and maritime Canada, is the largest lobster fishery in the world by volume and value (Wahle *et al.* 2020). The State of Maine accounted for 81% of national landings between 2018 and 2022, representing an average of \$506.5 million in ex-vessel value per year (106.1 million pounds) (NOAA 2022). The vast majority of these lobsters (80%) were landed in eastern Maine between Penobscot Bay and the Hague Line along the US–Canadian border (Fig. 1, Zones D–A) (ACCSP 2023), where coastal communities are characterized by lower

median income, education levels, and health care coverage than communities along the rest of the state's coastal region (Burnham et al. 2025).

In 2022, there were 5821 commercial lobster harvesters eligible to land their catch in Maine, including those with student and apprentice licenses (See Table S1 for a list of the different types of lobster licenses). Harvesters in Maine fish in one of seven geographic areas known as Maine Lobster Management Zones. A portion of harvesters also have federal permits that allow them to fish in adjacent federal Lobster Conservation Management Areas (LCM) (>3 nautical miles from shore) (Acheson 2003, Fig. 1). Nearly all Maine-based federal licenses fish in LCM Area 1. With some nuance, state and federal license holders are allowed to fish up to 800 traps. Each trap must have a trap "tag" affixed to it, which harvesters purchase from the Maine Department of Marine Resources (DMR) annually. The State of Maine reported that license holders purchased 2 790 942 trap tags in 2022, not including tags sold for recreational use. Although the fishing fleet does not purchase the maximum possible number of trap tags (4.6 million), the individual trap limit is high in comparison to other fixed gear lobster fisheries around the world, including in comparison to Canada where commercial trap limits for the same lobster species range from 100 to 400 per license (DFO 2008, 2021, 2022).

In the past, the majority of license holders in Maine have favored efforts to reduce trap limits (Holland 2011), but mounting regulatory pressure to reduce risk of whale entanglement has made this issue divisive (Kilchenmann et al. 2024). At the center of this divisiveness is debate about the extent to which the lobster fishery creates risk to NAWR recovery. Part of what has made this issue challenging is that there is limited information about fishing effort in the Maine lobster fishery, thereby making it difficult to evaluate the risk of the fishery on NAWR or how NAWR-related regulations could impact harvesters.

As context, NARW are a critically endangered large whale species that were historically overfished (Ross et al. 2021). Their population experienced a modest period of recovery in the early 2000's, but since around 2010, it has been in decline due to entanglements in fishing gear, vessel strikes, decline in calf numbers, changes in summer feeding areas, and changing biological patterns and environmental stressors caused by climate change (Meyer-Gutbrod et al. 2021, Ross et al. 2023). In an effort to protect the species, regulations implemented by the National Marine Fisheries Service (NMFS) (NMFS 2021) were modified in the aftermath of a court rule, which found that the lobster fishery was in violation of the US Endangered Species Act and Marine Mammal Protection Act (MMPA) (CBD v Ross 2018). The central issue driving this decision was concern that NARW are susceptible to being entangled in fixed fishing gear, including lobster traps and their associated ropes. Federal regulations introduced in 2021 sought to address this issue by reducing risk of serious injury and mortality to the whales by 60% through the creation of new seasonal closures in the region and by altering fishing gear (e.g. modified weak link insert requirements) and gear configurations (e.g. increasing the minimum number of traps per trawl to remove the number of lines in the water) (Taking of Marine Mammals 2022). These regulations, however, were initially deemed inadequate for reducing the risk of NARW mortality to an acceptable level within a 6-month period as mandated by the MMPA (CBD v Raimondo 2022), leading NMFS to increase the risk reduction target to 90% (NMFS 2023).

In pursuit of achieving the risk reduction target, NMFS relied on a Decision Support Tool (DST) to "evaluate management options that reduce the risk of entanglement of large whales in commercial fixed-gear fisheries under the jurisdiction of the Atlantic Large Whale Take Reduction Plan by allowing stakeholders to compare candidate management actions to a baseline level of risk" (Miller et al. 2024). The DST calculated risk to NARW as a function of the number of NARW per unit area relative to fishing effort. Specifically, the DST focused on "the amount of gear in the water, the duration of fishing, and the level of threat based on the gear configuration (i.e. rope diameter, net dimensions, and number of endlines) on a monthly basis." The relationship between "risk" and fishing activity placed fishing effort at the center of determining baseline risk and evaluating the effects that a management action would have on reducing risk. However, as stakeholders worked to identify an approach to meeting the risk-reduction target, two key dynamics changed the timeline for management action. First, congress used the US Consolidated Appropriations Act (2023) to halt any new NARW-related regulation in the fishery until 2028. A court decision by the US Court of Appeals for the District of Columbia Circuit then overturned the Biological Opinion, asserting that the assumptions that NOAA used to model risk were not defensible because it assumed a "worst-case" scenario.

While management decisions were ultimately delayed, the threat of possible NARW-related regulations coupled with the constellation of other stressors that have faced the lobster fishery has put immense pressure on the fleeting fleet. By making fishing effort the focus of our analysis, this review helps to address the existing data gap surrounding fishing effort, while simultaneously responding to more general calls for research about the socioeconomic dynamics of the lobster fishery in the midst of mounting stressors. Whether for NARW-related management actions or more general policy decisions, understanding fishing effort in the lobster fishery and how different segments of the fleet are or are not responding to different stressors is vital to making effective management decisions that account for the wellbeing, adaptive capacity, and resilience of the lobster fishery.

Data sources, cleaning, and preparation

Our analysis draws on multiple sources of confidential and non-confidential licensing, harvester, and dealer data spanning the 15-year period from 2008 to 2022 (Fig. 2). All data used in this study were cleaned and analyzed in R (Version 2023.09.1 + 494) using a tidy data approach (Silge and Robinson 2017). The decision criteria for cleaning data are reported in Table S2.

Non-confidential licensing data, stored in the Maine Department of Marine Resources' Licensing Enforcement and Environmental Data System (LEEDS), was used to identify individuals who hold lobster licenses and their demographic information (age and gender) and fishing location (management zone). There are 19 different lobster licenses that are issued by the state, based on harvesters' age, student status, or tribal affiliation (Table S1; 12 MRS1977). Licenses are further broken down by the scale of a harvester's lobster fishing operation and crew size (LC1 = captain only; LC2 = 1 crew member; and LC3 = 2 + crew members) (Table S1). There is no barrier for a license holder to switch between license types. For example, an LC1 or LC2 license holder can buy an LC3 license if

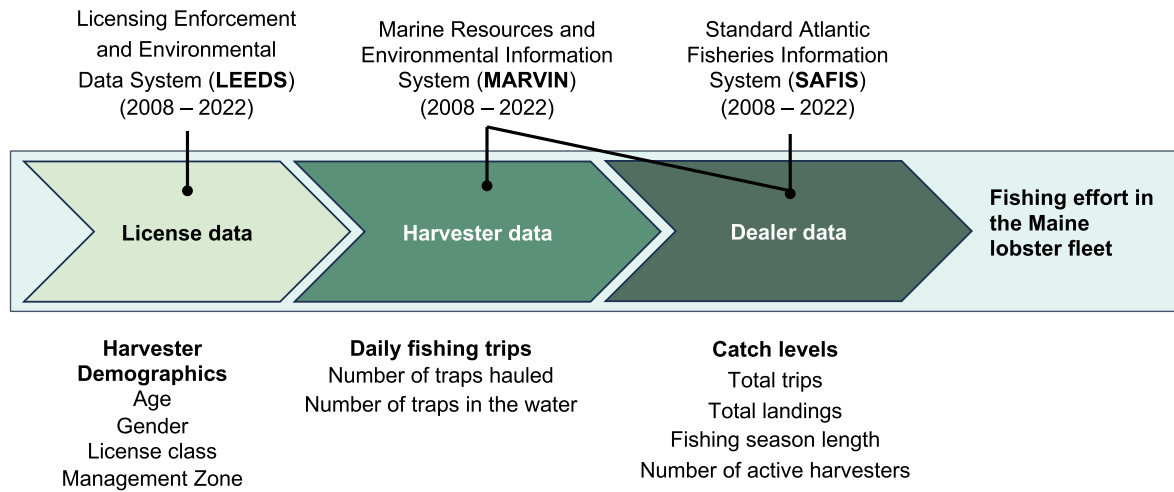


Figure 2. Data inputs for analysis of fishing characteristics, including license data (harvester demographics), harvester data (daily fishing trips), and dealer data (catch levels).

they want to carry more crew so they can haul more gear on a given day or fish more days. We limited our analysis to the nine most common lobster licenses, which collectively represented 98.6% of landings over the study period (see Table S2 for further rationale). These nine license types were binned by operational scale, resulting in four license groups for analysis (LC1, LC2, LC3, and student licenses (LCS) (Table S1). Since at least 51% of a harvester's traps must be fished in the management zone associated with their license (Fig. 1) and self-reported harvester data show that less than 1% of all fishing gear is deployed in other zones, we attributed all landings to the zone associated with harvesters' licenses.

Information on the number of traps hauled per day and the total number of traps a harvester has in the water was derived from confidential harvester logbook data held in the Marine Resources and Environmental Information System (MARVIN) database. Harvester logbooks include data on reported daily fishing trips for a selected subset of the Maine lobster fleet, including the number of traps fished and the total number of traps they have in the water (see ASMFC 2017 for a full description of the sampling method and how it has changed). The subset is randomly selected each year, without replacement. Over the study period, the active harvester report records included between 6% and 18% of the total active fleet. Confidential dealer reports, queried from the Standard Atlantic Fisheries Information System (SAFIS) and MARVIN databases, account for 100% of all harvester-to-dealer transactions for lobsters caught in both state and federal waters were used for information about the total trips, number of active harvesters, fishing season length, and catch. Finally, the number of lobster trap tags sold was downloaded from the Maine DMR website (DMR 2023).

Understanding fishing effort and how it has changed in the Maine lobster fishery

Fishing effort was analyzed using multiple individual and combinatory metrics (Table 1). By combinatory, we mean composite metrics that integrate multiple measurements of fishing effort together to produce a compound or derivative metric. Our decision to evaluate fishing effort in multiple ways recognizes that there are trade-offs associated with different metrics of fishing effort, including, in the case of the lobster

Table 1. Fishing effort metrics analyzed, type of metric, and the socio-demographic characteristics associated with them based on data availability.

Fishing effort metric	Metric type		Socio-demographic characteristic		
	Single	Combinatory	Age	Gender	License Class
Access	✓		✓	✓	✓
Participation	✓		✓	✓	✓
Harvester latency (%)		✓	✓	✓	✓
Fishing season length	✓		✓	✓	✓
Trap tags purchased	✓				✓
Traps hauled per day	✓				✓
Trips fished	✓				✓
Trap latency (%)		✓			✓
Catch per unit effort (CPUE)		✓			✓
Effort per unit catch (EPUC)		✓			✓

fishery, the ability to delimit different socio-demographic characteristics of the fleet (Table 1; see Socio-demographic characteristic columns), which is vital to studying resilience and adaptive capacity at different scales (Räsänen *et al.* 2016). We discuss each metric in the subsections below.

Access, participation, and latency in the lobster fishery

The first suite of fishing effort metrics that we considered related to access, participation, and latency in the lobster fishery. Access refers to the bundle of privileges conferred by a fishing permit or license as opposed to the act of fishing itself (Silver and Stoll 2022). Access is often associated with topics, such as harvester identity, sovereignty, and adaptive capacity (Silver and Stoll 2019, Bennett *et al.* 2021), but it is not a reliable metric for determining direct engagement in fisheries since an individual may hold a license latently or lease a license to another individual or corporation (Szymkowiak and Himes-Cornell 2015). Active participation, as defined by those who participate in the act of harvesting, is a reflection

of who is directly deriving benefits from a fishery (Beaudreau et al. 2019), rather than an indicator of the latent potential of those who could fish. Harvester access and active participation metrics can be integrated to determine the proportion of a given fleet that is inactive or latent in a given timeframe (Cucuzza 2020). Prior research related to the socioeconomic dimensions of the Maine lobster fishery has tended to focus on access or participation, but rarely together, while limited to no attention has been given to latency. To calculate harvester access, participation, and latency, licensing data were used to determine the number of eligible participants (ι) in the lobster fishery by aggregating all nine types of lobster licenses (a) across all zones (z) at an annual timeframe (Table S1). Dealer data were then used to identify active harvesters as defined as those who had at least one reported catch record in the dealer database (i). Active participation was analyzed at the month and year timeframe (Table S2). The following equations were used to calculate access, active participation, and % latency:

$$\begin{aligned}\text{Access} &= \sum \iota_{az} \\ \text{Active participation} &= \sum \iota_{iz}.\end{aligned}$$

Fishing season length

To augment our analysis of access, participation, and latency, we analyzed the length of harvesters' fishing season. Fishing season length has long been associated with fishing effort (Roy 1998), as the length of a fishing season is often viewed as a key determinant of the ability for harvesters to go fishing. This perspective is commonly institutionalized in fisheries management through input control measures such as closed seasons (e.g. Murawski et al. 2000) and days at sea (e.g. Lee 2012).

% latency = $\frac{\sum \iota_{az} - \sum \iota_{iz}}{\sum \iota_{az}} 012$ that aim to limit fishing effort by limiting the amount of time an individual is on the water. As a metric of fishing effort in fixed gear fisheries, fishing season length is useful in understanding the total amount of time gear is being fished over a given time period and when. While it does not capture several key dimensions of fishing effort (time on water, number of trips, and soak time) and would not reveal fishing phenomena like the "race to fish" (Smith 2012), fishing season length is especially salient to the lobster fishery in Maine because the total length of time gear is in the water and in what timeframe drives NARW entanglement risk. Furthermore, it is known that fishing effort is concentrated in the late summer and early fall to target the peak period for lobster molting when lobsters that had previously been too small to keep become legal (Holland 2011). Prior research related to season length has assumed that participants in the lobster fishery fish 365 days a year because the fishery is open all year (Myers and Moore 2020). To empirically evaluate the length of time, harvesters had gear in the water, we used dealer reports to calculate season length. We defined season length as the number of days between the first and last date that a harvester reported landings. Based on our knowledge of the fishery, this approach was taken because once harvesters put their gear in the water at the start of their season, they do not typically remove it from the water until the end of their season because it is a time-intensive and financially costly activity. Fishing season length was calculated as

$$\text{Fishing season length} = \frac{\sum_i (D\alpha - D\sigma)}{(\sum \iota_{az} - \sum \iota_{iz})},$$

where D represents the sequential day number of the calendar year, where 1 is the first day of January, α represents the last day that a harvester reported landings, and σ denotes the first day.

Tags, traps, and trips

To complement our analysis of season length, we evaluated trap tag purchases, traps hauled, and trips taken. In the same way that the number of lobster licenses is a reflection of the fleet's access potential, so too are trap tags purchases because the number of trap tags purchased is the theoretical maximum number of traps that can be in the water at any given point in the fishing year. In contrast, the number of traps fished represents the amount of gear in the water at a given point in time. Together with fishing season length, traps fished is an important metric for discerning the overall risk the fleeing fleet poses to NARW. Finally, the number of trips taken is important from a socioeconomic perspective because it denotes how often harvesters are checking their gear. This information helps to contextualize combinatory metrics such as catch per unit effort (CPUE).

Catch per unit effort

The above metrics of fishing effort were augmented by calculating CPUE, a common metric used to evaluate catchability and resource abundance (Mauder and Punt 2004, Hoyle et al. 2024). While CPUE is historically used for stock assessment purposes, CPUE is a combinatory metric that is inherently tied to fleet dynamics and profitability. Previous research by Boenish and Chen (2018) calculated CPUE in the Maine lobster fishery (2008–2013) using data from a sea sampling survey run by the Maine Department of Marine Resources. We calculate CPUE by dividing total lobster landings (in pounds) by total effort as a function of the number of traps fished in a given time period. Total landings were summarized from daily dealer reports and total effort was extrapolated from harvester logs (see below for details of the extrapolation method). CPUE was calculated by zone at an annual temporal scale and by month across the entire state (Fig. 7).

$$\text{Catch per unit effort} = \frac{\sum \text{landings}}{\left(\frac{1}{\delta/\alpha} * \left(1 - \frac{(\sum \iota_{az} - \sum \iota_{iz})}{\sum \iota_{az}}\right) * \sum t\right)},$$

where δ is the total number of harvester records and t is the number of traps fished.

The final step in calculating CPUE was to extrapolate the results across the entire fishing fleet. The harvester logbook dataset includes information from daily fishing trips that could be used to calculate fishing effort from a subset of the overall lobster fleet in any given year (h) (see "Data sources, cleaning, and preparation" section). Because each yearly subsample is random, extrapolating data from the harvester dataset must account for the heterogeneity of the fleet based on license classes. To do this, we calculated a distinct effort multiplier for each year, month, and license class. The effort multiplier was calculated by dividing the total number of active harvesters (those with a record in the dealer data) by the total number of harvesters that submitted a harvester log. Due to the relatively low number of student license holders, there were months when none were represented in the harvester data. In these cases, we assigned the total number of active student harvesters as the effort multiplier. Ultimately, we used the effort multiplier to extrapolate the number of traps

fished and the total number of traps in the water recorded by year, month, and license type. The resulting values provide an estimate of these metrics for the entire fleet.

$$\text{Effort multiplier} = \frac{(\sum t_{az} - \sum t_{iz})}{\sum t_{bz}}.$$

The extrapolation method described assumes that all harvesters of the same license-type fish the same number of traps and have the same number of traps in the water. Although this assumption has the potential to either overestimate or underestimate the extrapolated outputs, we are confident that the extrapolation method is generally representative of the entire fleet because we can compare extrapolated landings to observed landings from the dealer data. When we conducted a simple *t*-test on the annual landings from the extrapolated and observed data, we found no significant difference between the means ($t = -0.48$, $P = 0.63$).

Effort per unit catch

In addition to calculating CPUE, we also calculated effort per unit catch (EPUC). EPUC is not a common metric in the fisheries literature in general, but past research on the Maine lobster fishery has used it to evaluate the Maine lobster fishery's "overcapacity" relative to Lobster Fishing Area 34 (LFA 34) in Canada (Myers and Moore 2020). Specifically, Myers and Moore 2020 used estimates of fishing season and number of traps fished to calculate EPUC in the Maine lobster fishery and then compared these finds to the EPUC in Lobster Fishing Area 34 (LFA 34) in Canada to determine the relative effort or the extent to which EPUC differed between the two adjacent areas. The results of that 2020 study found that the Maine lobster fishery expended 7.5 times more effort to land the same amount of catch. To compare our results to their findings, we re-calculated EPUC in Maine and LFA 34 using empirical licensing, harvester, and dealer data rather than proxy data for the original timeframe of the previous study (2016–2017) and extended it forward and backward from 2008 to 2022. The calculation for EPUC is the inverse of CPUE and relative effort is calculated by dividing EPUC in Maine by EPUC in LFA 34. While Myers and Moore (2020) previously used the term "overcapacity," we adopt the term "relative effort" here to avoid any suggestion that LFA 34 is operating at a more ideal capacity.

Results

Fleet-wide dynamics

From shifting settlement patterns, decreasing productivity, and declining bait availability to market swings, global trade disruptions, and gear conflict with NAWR, participants in the lobster fishery have faced a diverse suite of stressors. Yet over the last decade and a half, fishing effort at the fleet level has been remarkably stable. Across both the individual and combinatory metrics, we observe a pattern of stability. Access to the lobster fishery, as defined by the number of license holders per year (all types combined—Table S1), was relatively stable during the 15-year study period (mean = 5 698, SD = 135.7) (Table S3) (Fig. 3a). Like access, participation and latency in the lobster fishery also showed relative stability at the fleet level. During most of the 15-year study period, 71%–74% of license holders were active in any given year. The only exceptions occurred in 2020 when active participation dipped to 69% and in 2022 when we observed a period-low in par-

ticipation (67%). The lowest mean season length across the fleet also occurred in 2020, overlapping with the start of the global COVID-19 pandemic. However, 2020 was an exception to an otherwise stable season length over the study period (mean = 200.74 days; SD = 4.75 days) (Fig. 4a). Similarly, the total number of annual trips has been stable, except for 2020 and 2022, when there were observed dips. We also see stability in the number of lobster trap tags sold relatively to the proportion of traps being fished (Fig. 6a). On average, the lobster fleet purchased 2 943 837 trap tags per year. The most tags ($n = 3\ 160\ 602$) issued during the study period occurred in 2008 and then sales gradually decreased. Similarly, the total number of traps hauled during the study period was relatively stable (mean = 56 540 223; SD = 4 227 818), though annual traps hauled also dropped in 2020 and 2022 along with the number of trips (Figs 5b and 7b). Mean CPUE in the lobster fishery over the 15-year study period was 1.9 pounds per trap (SD = 0.28) (Fig. S4). CPUE increased from 2008 to 2012 and then remained steady over the subsequent 11 years. CPUE has a strong seasonal pattern to it, which has been stable through time. CPUE is lowest from March through June (mean = 0.89). CPUE then begins to increase in July (mean = 1.76), though this increase is somewhat variable through time (SD = 0.53), reflecting the year-over-year variability in the timing of the summer molt (see Mills et al. 2017, Staples et al. 2019) (Fig. 7b). CPUE continues to increase through the summer and fall, reaching a peak in October (mean = 2.36) and December (mean = 2.38). Mean EPUC in Maine over the study period was 0.54 and in Canada's LFA 34, it was 0.41 (Fig. 7d). EPUC in both Maine and LFA 34 has slightly decreased over the 15-year study period and reached a near-low in 2022. Relative capacity in the Maine lobster fishery is 1.31 times higher than LFA 34, but has not changed much through time. Despite the semblance of stability at the fleet-wide scale, we observe a number of large-scale structural changes occurring within the fleet. In the subsections that follow, we discuss each of the focus fishing effort metrics and how they have changed through time within different facets of the fleet.

Intra-fleet dynamics

Access, participation, and latency in the lobster fishery

Despite fleet-wide stability in access, participation, and latency, we observe intra-fleet variability in each of these metrics. While the ratio of male to female license holders remained relatively stable, the absolute number of female license holders increased (on average, 93% of license holders self-identify as male, but the number of female harvesters increased by 41%) (Fig. 3b). By age, across the study period (2008–2022), the total number of licenses issued to those between 18 and 45 (–14%) and 46 and 69 declined (–12%), while the number of licenses issued to those younger than 18 (18%) and over 70 increased (48%) (Fig. 3c). Separating harvesters' access, participation, and latency by license class, further illuminates the intra-fleet changes that have occurred in the fishery. Most striking is the decline (–39%) of Class 2 licenses and the increase of Class 3 licenses (104%) (Fig. 3d). There was also a notable geographic change in the distribution of licenses. Licenses became more concentrated in the eastern part of Maine (Table S3, Fig. S1). Zones D, C, and A alone accounted for 60% of the licenses in 2022 ($n = 3383$) (Table S3, Fig. S1) as a result of the decline in number of licenses in the western



Access, participation, and latency, however, are not perfectly correlated. Across the study period, latency is much higher among female participants (50%) (Fig. 3b), younger (<18) (45%), and older harvesters (+70) (37%) than other groups (Fig. 3c). Latency also differs by license class as those

Season length differed greatly by gender (female = 96.2 days; male = 206.0 days), age (sub-18 = 90.8 days; 71+ = 149.0

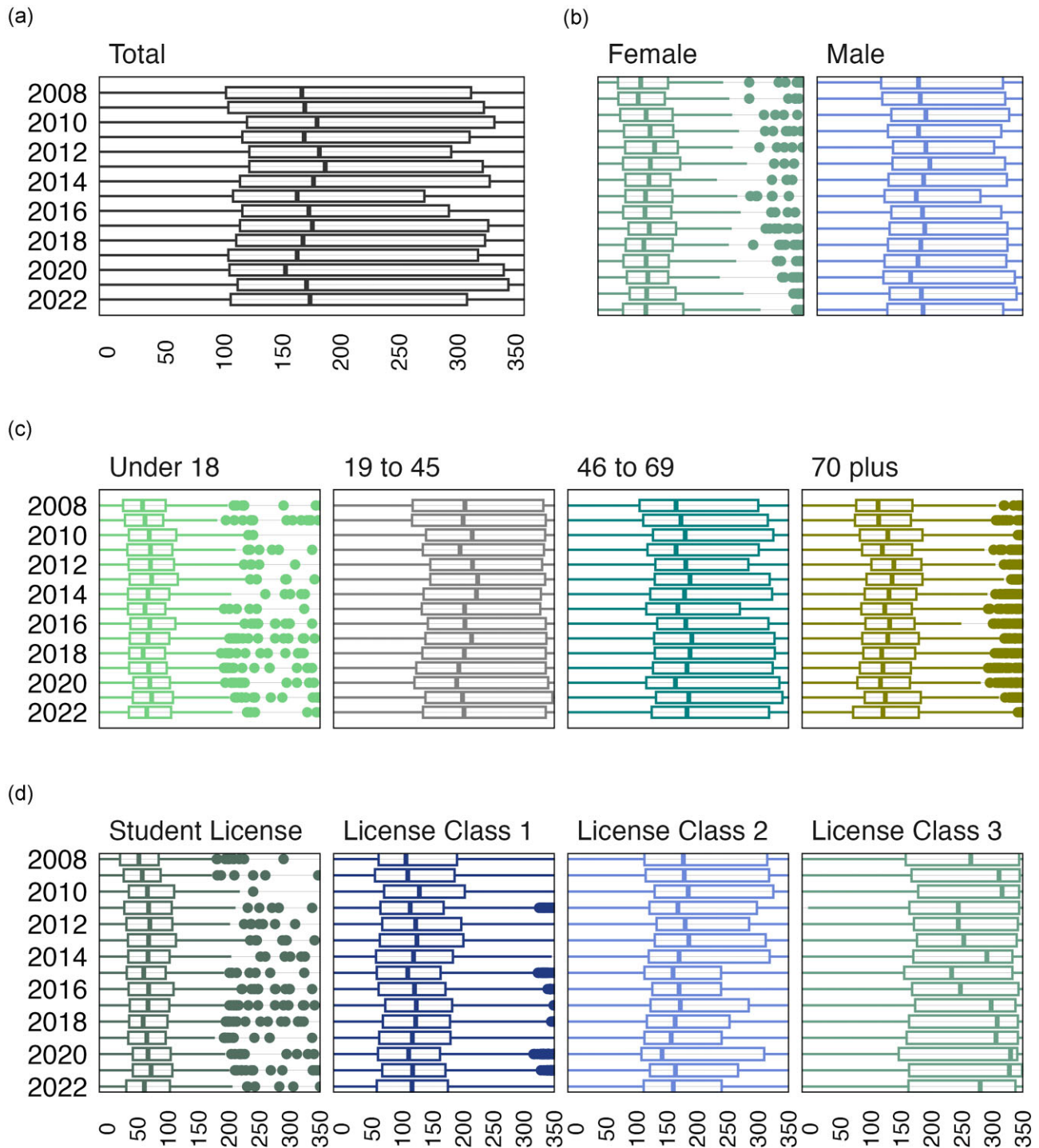


Figure 4. Distributions of average length of season (days) that harvesters have lobster fishing gear in the water for (a) the entire fleet, (b) by gender, (c) by age group, and (d) by license class from 2008 to 2022.

days), and license types (~50 days separated each license type) (Fig. 4b–d). Those under 18, students, and female-license holders had the shortest observed fishing season lengths and appear to target fishing in the summer months when weather conditions are best, CPUE is high, and school is out of session. Those with Class 3 license classes reported the longest fishing seasons.

Fishing seasons in the lobster fishery exhibited a strong seasonal trend with the peak (68% of trips) occurring from July

to October that has not changed through time (Fig. 5a–c). This pattern is similar across gender, age, and license type, but the extent to which different subgroups of harvesters participate in the fishery in the other months differs. The most fishing trips observed during any month ($n = 54\,641$) occurred in August 2017 (Fig. S2). Between 2008 and 2022, the number of trips per year across the fleet has decreased (–21%) from 252 357 to 199 156 trips, with a peak in 2016 ($n = 275\,000$) (Fig. 5b). This overall decline has been driven by a dip in

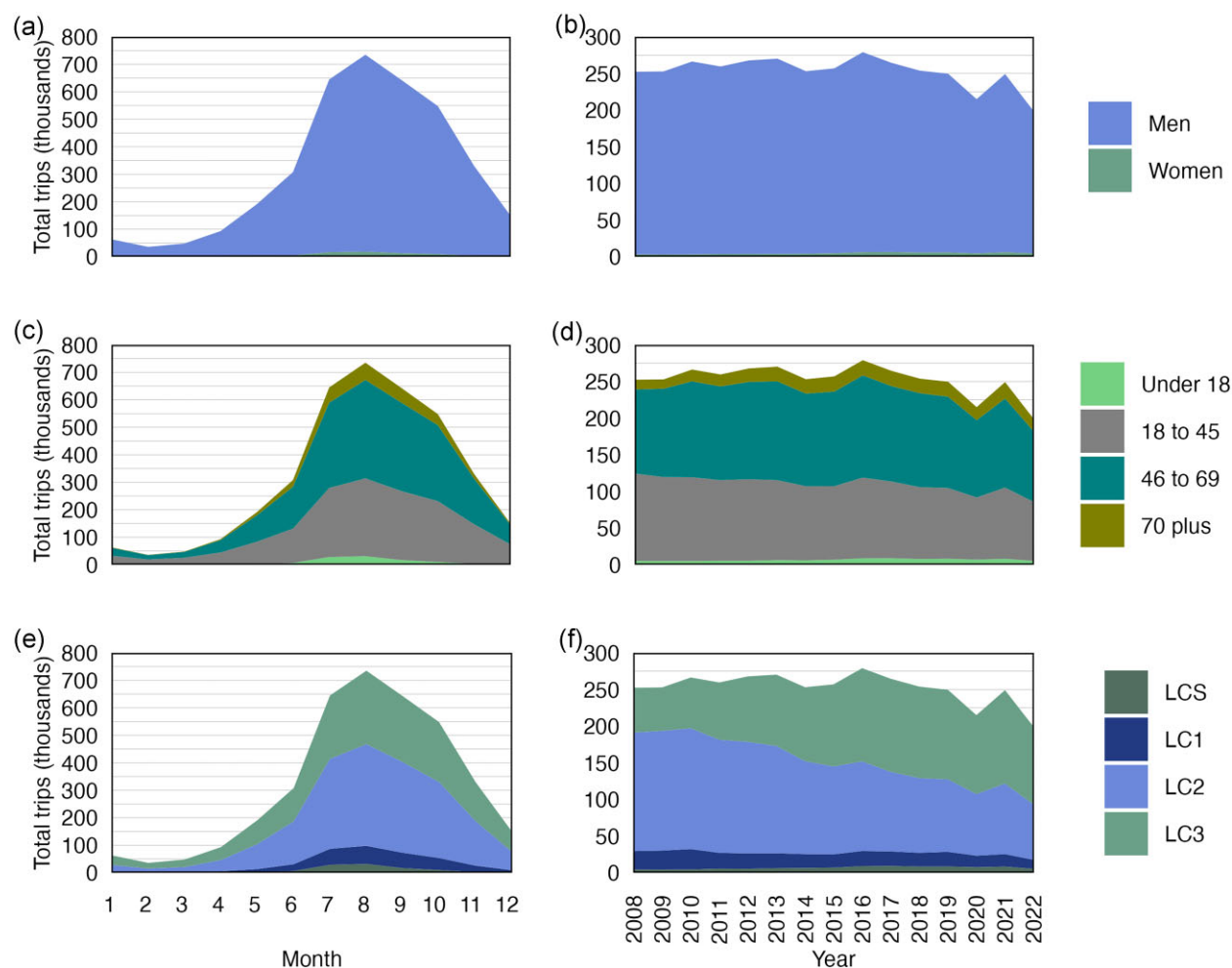


Figure 5. The total number of annual trips taken from 2008 to 2022 by commercial lobstermen by (a) gender, (c) age group, and (e) license class and the total number of trips taken per month across 2008–2022 by (b) gender, (d) age group, and (f) license class.

trips since 2019, particularly among Class 1 (−51.3%) and Class 2 (−53.4%) license holders (Fig. 5b). Within the fleet, we also observe a change in trips by license type, with the proportion of larger operations (Class 3) increasing (+15%) relative to the proportion of smaller operations (Class 1 and Class 2) (−20%) (Fig. 5b). There is also a regional trend to decline in trips. Although trips have decreased in all zones, the observed decline is most prominent in Zones D, E, F, and G where trips have declined by 32%, 37%, 32%, and 20%, respectively. In contrast, trips in the eastern portion of the state (Zones A, B, and C) have declined by 16%, 6%, and 8%, respectively.

Further inspecting the fleet-wide decline in the number of trap tags purchased reveals that decreases are driven by a relatively sharp decline in traps associated with Class 2 licenses (Fig. 4b) and Zones F and D. The fewest trap tags issued by count number were in Zones G, E, and B, which correspond to the zones with the fewest licenses. We estimate that 61% (2008 and 2022) to 71% (2017) of traps sold are fished, indicating that trap tag sales are a useful indicator of potential effort, but not actual effort on the water. Socio-demographic data associated with trap sales are not available, so it was not possible to separate trap tags sold by age or gender, but differences in license class are evident. Most notably, the number of traps actively fished among Class 2 license holders declined

(−58%), while the number of them sold and fished among Class 3 license holders increased (+123%).

Catch, Traps Hauled, Catch Per Unit Effort, and Effort Per Unit Catch

Similar to trap tags, we also observed that the fleet-wide decline in lobster landings was driven by a decline in the total proportion of catch among Class 2 license holders (−37.3%) (Fig. 7a). Total landings by Class 3 license holders, in contrast, increased (+183%) from 2016 to 2022, then decreased by Class 3 license holders decreased by 19.0%. There was also a large decline in the total number of traps fished by Class 2 license holders, while the number of traps fished by Class 3 license holders has increased (Fig. 7b).

Within the fleet, CPUE could only be calculated at the zone and license class level because there are representation issues when trying to extrapolate harvester data by age and gender (Fig. 7c). CPUE was consistently lower in the southwestern region of the coast (Zones G–E; mean = 1.4, SD = 0.24) relative to eastern Maine (Zones D–A; mean = 2.1; SD = 0.47). Zone B had the highest mean CPUE over the study period, peaking at 3.8 pounds per trap in 2022. CPUE is highest among Class 3 and more than twice as high as student license holders (Fig. 7c). Like CPUE, EPUC calculations were also constrained to

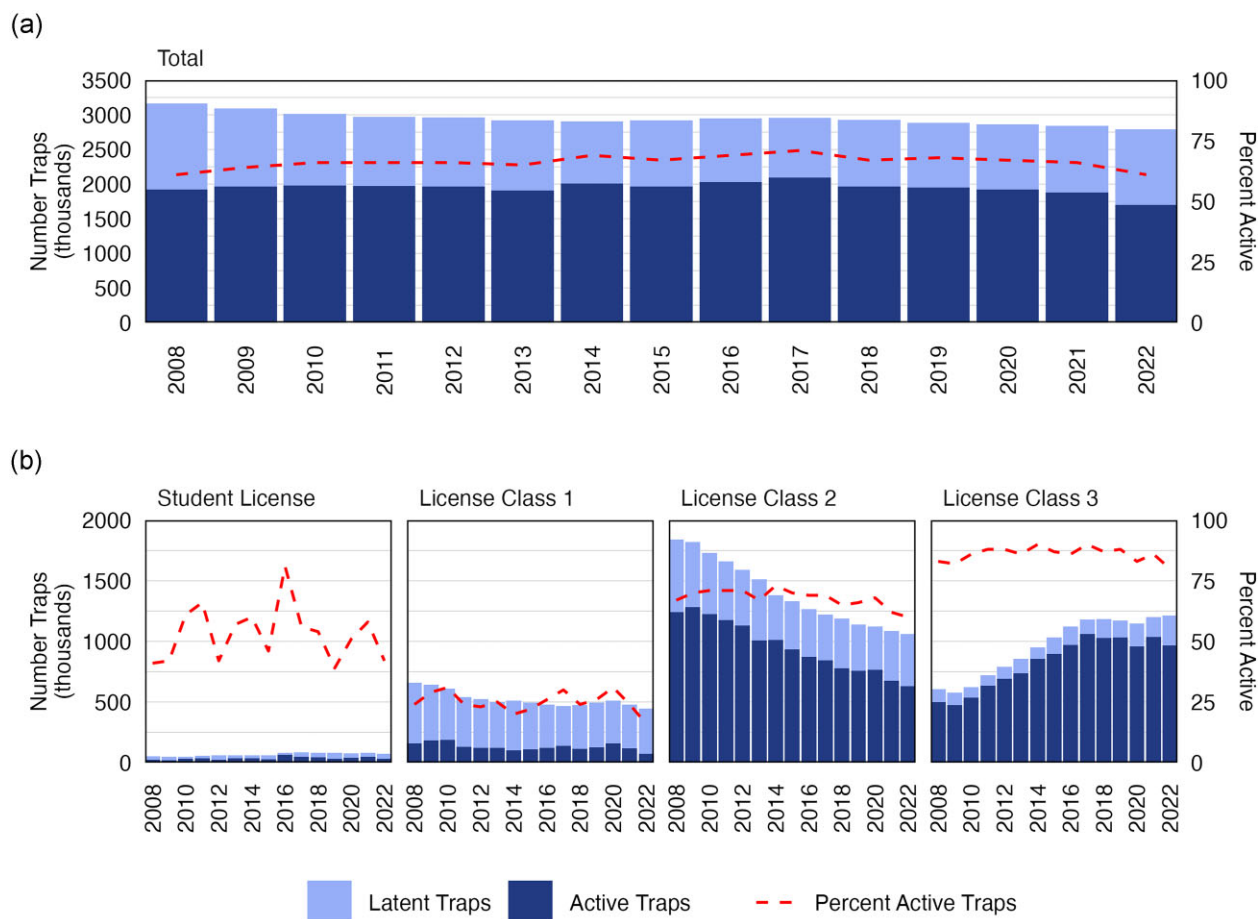


Figure 6. Number of lobster trap tags sold compared to the number of traps fished for (a) the entire fleet and by (b) license class from 2008 to 2022. Total lobster traps fished based on extrapolated data. The dotted line denotes the percentage of trap tags fished.

focusing on license class levels within the fleet. EPUC is lowest among Class 3 licenses and most similar to EPUC measurements in Canada's LFA 34.

Discussion and conclusions

Known for its unique co-management system and history of collective action (Wilson *et al.* 1994, Acheson and Gardner 2014), the lobster fishery in Maine has emerged as one of the most sociocultural and economically valuable fisheries in the world (Billings 2014, Wahle *et al.* 2020). However, after almost two decades of near-continuous growth, it appears the fishery resource may be in decline (McManus *et al.* 2023). Concurrent with the apparent change in resource availability, the fishing industry has also been confronted by multiple other challenges that are threatening its long-term resilience. At the forefront of these challenges are future policy changes to protect NARW. Amidst this period of mounting uncertainty in the lobster fishery, the lack of socioeconomic information about the fishery, including baseline information about fishing effort, makes it difficult for decision makers to evaluate the current state of the fleet or how it is adapting to stressors (ASMFC 2020). Aiming to help address this gap and responding to calls for long-term and longitudinal approaches to monitoring social-ecological resilience and vulnerability (Fawcett *et al.* 2017), our analysis provides a review of fishing effort

in the Maine lobster fishery over a 15-year period, examining both fleet-wide patterns and intra-fleet dynamics.

At the fleet-wide scale, fishing effort in the lobster fishery has been remarkably stable, especially given the environmental and socioeconomic stressors confronting the industry. This appearance of stability is reflected in many of the fishing effort metrics reported at the aggregate level, including fisheries access, participation, season length, total trips, traps fished, catch per unit and effort, and effort per unit fished. To make sense of system-level patterns like these, scholars have argued that it is critical to consider intra-system dynamics (Manson 2001, Dewulf *et al.* 2019) as subsystem changes can provide early evidence of vulnerability and often precipitate social-ecological regime shifts (Adger 2006, IPCC 2007).

The intra-fleet changes observed in the lobster fishery stand in direct contrast to the observed fleet-wide stability. Change within the lobster fleet was particularly evident across age, scale of operation, and geography, suggesting several potential vulnerabilities within the fishery. Three intra-fleet dynamics warrant particular emphasis. First, while overall access to the fishery has remained stable, the number of license holders between 18 and 69 years of age has declined. In contrast, those under 18 have remained steady, while harvesters over 70 have increased. This intra-fleet shift in demographics is important from a resilience perspective because it coincides with the age of harvesters who are in the prime of their professional



carers. This finding is consistent with prior research that describes the graying of the fleet in fisheries in Maine (Johnson and Mazur 2018) and elsewhere (Donkersloot and Carothers 2016). A decline in this subpopulation of harvesters is concerning because this suggests that the core workforce is diminishing even if the total number of individuals is not. Not only does this have implications for overall production in the fishery, but it also may negatively impact participation in the management system as this group is also mostly likely to engage in the governance process since they are most invested in the fishery (Farr et al. 2018, Foley et al. 2020). Second, beyond the shift in demographics, there has also been a near-transformation in the structure of the fleet as defined by scale of fishing operation. Early in the study period, Maine's lobster fleet was dominated by Class 2 license holders (48%). Fifteen years later, only 30% of the licenses were Class 2 license holders and the ratio of Class 2 to Class 3 licenses in the fishery became 1.07:1.00 as the number of Class 3 license holders proliferated. This intra-fleet change is indicative of a shift within the lobster fleet toward larger fishing operations that are capable of carrying more crew and engaging in the fishery for a longer period of time each year. While the analysis presented in this paper does not focus on explaining why this shift has occurred, our observation from engaging with the fishing fleet is that the shift is being driven by real or perceived perceptions that larger operations are increasingly as-

Examining intra-fleet dynamics also brings attention to the heterogeneity of harvesters in the lobster industry and how they engage in the fishery. Findings from this review complement scholarship by Dayton and Tokanaga (2023) by highlighting differences in fishing effort based on socio-demographic characteristics such as age and gender as well as license type. For example, differences in season length among license holders varies between Class 3 license holders

(mean = 265 days) and Class 1 license holders (mean = 146 days). Differences like these will invariably shape the extent to which future changes in the fishery impact subsets of harvesters as well as their capacity to adapt. Even in the analysis presented here, there is evidence that different groups appear to be emerging as possible “winners” in the face of recent changes. Changing heterogeneity within the fleet will likely affect the impact that these pressures will have on different fishery participant groups, as well as their capacity to adapt.

In addition to bringing attention to the heterogeneity of fleet dynamics, this paper also highlights the importance of continued efforts to understand fishing fleet dynamics and how they differ among segments of the fishery. Specifically, the results presented in this paper help to refine prior estimates of a key metric of fishing effort in the Maine lobster fishery (see Myers and Moore 2020). This latter insight bears significance because the lobster industry is under mounting pressure to reduce risk of gear interactions with large marine mammals, and future management decisions will likely hinge on estimates about fishing effort and the probability of marine mammal interactions. The difference between the results from this analysis and prior research stems from prior assumptions about fishing effort. Although the lobster fishery in Maine is open year-round (365 days) and harvesters, with some exceptions, are able to fish up to 800 traps, effort across the fleet is well below both the theoretical maximum and prior estimates. Three interrelated factors contributed to this observed disconnect. First, the fishery was considerably more seasonal than regulations dictate, as 67% of fishing trips occur during the summer and fall between July and October over the last 15 years. Second, latency in the fishery was high. In 2022, 33% of the 5821 license holders across the fleet did not participate in the fishery. Third, the total number of fishing trips per year decreased among active participants (−21%). Taken together, there were fewer active harvesters, taking fewer trips, in a relatively condensed fishing season compared to the fleet’s potential, and overall fishing trips declined. These results also help to contextualize the Maine lobster fleet’s fishing effort relative to the adjacent lobster fishery in Canada’s LFA. Results from this study show that across the study period, the average EPUC in Maine was 0.54, while in LFA 34, it was 0.41. This finding confirms that Maine’s fishery is operating less efficiently than that in LFA 34, but the relative capacity metric is only 1.31. This result is 17 times lower than what was reported in the literature previously. Given that EPUC is highest among student license holders in Maine and that this group of individuals is vital to sustaining participation in the fishery, careful consideration of the socioeconomic impacts should be considered before using fleet-wide relative capacity metrics to guide management decisions.

Finally, it should be noted that although there was fleet-wide stability for most of the study period, there has been an observable decrease in fishing effort since 2020. Landings in Maine in 2022 are down 26% from their peak in 2016 and the overall value of the catch is 48% lower than it was in 2021 when the value of the fishery reached \$742.9 million (DMR 2024). Trips were also down by 27% and the estimated number of trips fishing reached the lowest level in the study period in 2022. The extent to which the Maine lobster fleet is able to be resilient in the face of continued pressures and change remains a question that warrants close monitoring.

Future research on fleet dynamics in the Maine lobster fishery would benefit from consideration of several limitations associated with the analyses presented in this paper. First, the analysis relied on an extrapolation of harvester data across the entire fleet. While we take care to minimize potential extrapolation bias (see method section), it is important to recognize that the CPUE calculations are estimates. Moving forward, new requirements for 100% harvester reporting in the lobster fishery starting in 2023 will allow researchers to more accurately calculate and characterize CPUE of the fleet. It is also important to acknowledge that the state’s sampling protocols for harvesters changed through time, potentially introducing bias. Specifically, from 2008 to 2018, the state randomly selected 10% of license holders from each license type/zone, including inactive harvesters, and then in 2019 used an optimized selection process that prioritized active participants. Further, it is necessary to acknowledge that our analysis of CPUE does not include data from annually selected harvesters (as part of the 10%), who are required to report through federal Vessel Trip Reporting due to data access challenges. The absence of this data represents 3%–4% of harvesters since 2019 and, while we suspect it has a nominal effect on our results at the scale that we report CPUE in this paper, its absence would pose an issue for analyses at finer spatial and temporal scales (especially offshore during winter months when overall effort is low). Finally, we also acknowledge that tribal licenses were not included in our analysis because the state does not maintain a complete record of these data. This is a complicated issue related to data sovereignty and fishing rights that is well beyond the scope of this paper, but the absence of these data means that tribal participation in the analysis is not visible. Despite these limitations, this paper advances efforts to understand fishing fleet dynamics and how they differ among segments of the fishery. Continued attention to the human dimensions of fisheries is vital to making well-informed policy decisions in the face of change, including the iconic Maine lobster fishery.

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Author contributions

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